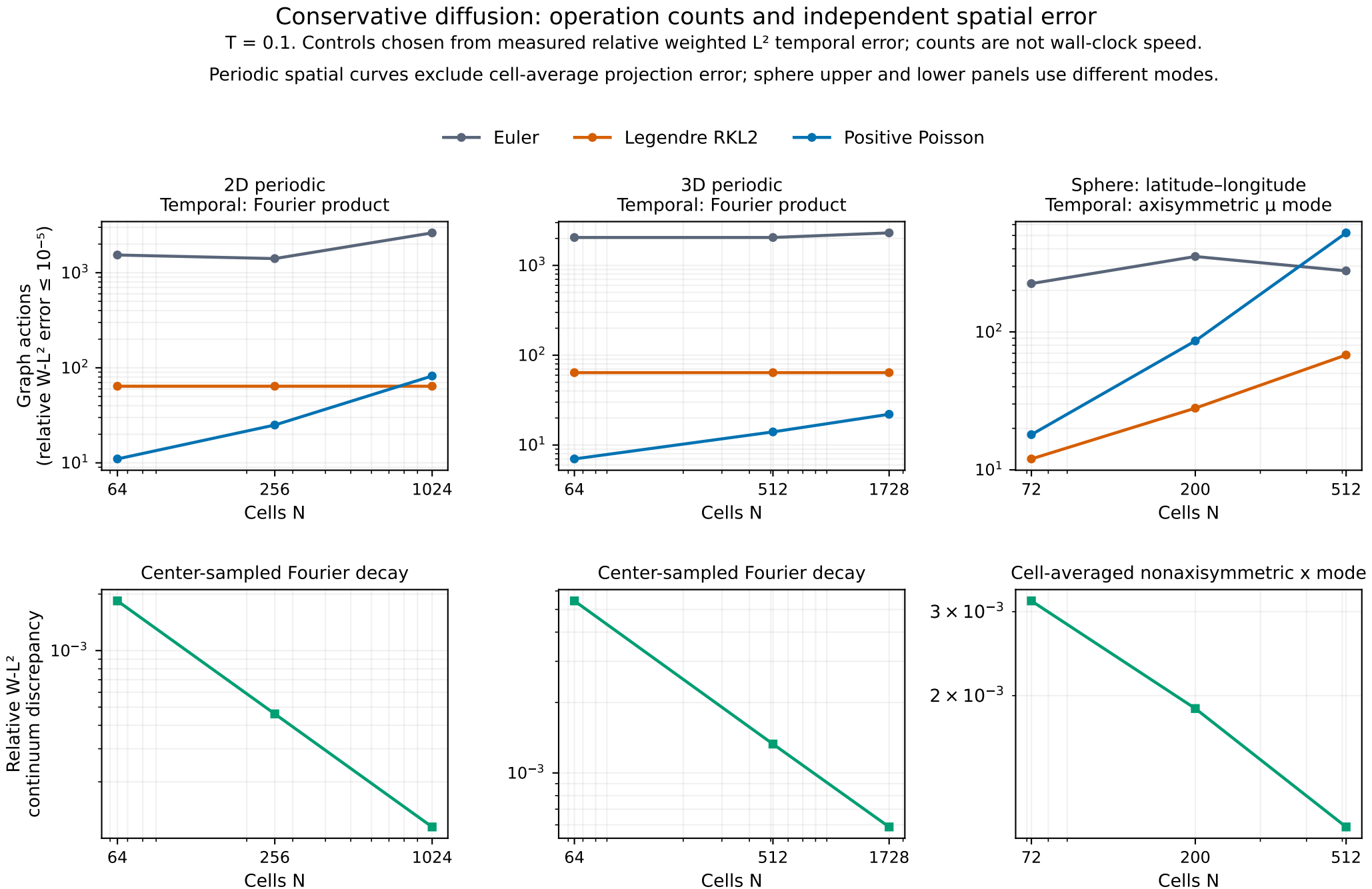
Conservative diffusion: operation counts and independent spatial error
T = 0.1. Controls chosen from measured relative weighted L² temporal error; counts are not wall-clock speed.
Periodic spatial curves exclude cell-average projection error; sphere upper and lower panels use different modes.
Euler
Legendre RKL2
Positive Poisson
2D periodic
Temporal: Fourier product
3D periodic
Temporal: Fourier product
Sphere: latitude–longitude
Temporal: axisymmetric μ mode
Graph actions
(relative W-L² error ≤ 10⁻⁵)
Cells N
Center-sampled Fourier decay
Center-sampled Fourier decay
Cell-averaged nonaxisymmetric x mode
Relative W-L²
continuum discrepancy